

Homework 01

Question 1 (Least squares and matrix calculations)

Set the random seed to 43201. Generate $n = 100$ observations with $p = 5$ predictors according to

$$x_{ij} \stackrel{\text{iid}}{\sim} N(0, 1), \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, 0.7^2),$$

with all predictors and errors generated independently. Let

$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_5], \quad \boldsymbol{\beta} = (3, 1.4, -0.9, 0.7, 0, 1.1)^\top,$$

and generate

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

- Generate the data and report the dimensions and rank of $\mathbf{X}$. Calculate the least-squares estimate $\hat{\boldsymbol{\beta}}$ using a numerically stable least-squares routine, without explicitly calculating $(\mathbf{X}^\top \mathbf{X})^{-1}$. Compare the estimates with the coefficients used to generate the data.

```
set.seed(43201)
n <- 100
p <- 5
X <- matrix(rnorm(n * p), n, p)
X <- cbind(1, X)
beta <- c(3, 1.4, -0.9, 0.7, 0, 1.1)
epsilon <- rnorm(n, 0, 0.7)
y <- X %*% beta + epsilon
dim(X)
```

```
[1] 100  6
```

```
qr(X)$rank
```

```
[1] 6
```

```
betahat <- qr.solve(X, y)
z <- cbind(beta, betahat)
z
```

```

      beta
[1,]  3.0  3.0063210
[2,]  1.4  1.4171604
[3,] -0.9 -0.8883453
[4,]  0.7  0.7952834
[5,]  0.0  0.0484378
[6,]  1.1  1.1319341

```

As can be seen, the least-squares estimates are very close to the true coefficients used to generate the data. We also found that the dimensions of $\mathbf{X}$ are 100×6 and its rank is 6, indicating that the columns of $\mathbf{X}$ are linearly independent.

- b. Calculate the fitted values, the residual vector $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\beta}$, and the training root mean squared error

$$\text{RMSE} = \left(\frac{1}{n} \sum_{i=1}^n r_i^2 \right)^{1/2}.$$

```

yhat = X %*% betahat
r = y - yhat
rmse = sqrt((1/n) * sum(r^2))
rmse

```

```
[1] 0.6763696
```

- c. Report $\|\mathbf{X}^T \mathbf{r}\|_\infty$. Explain why this value should be close to zero and what it tells us about the residual vector.

```
norm(t(X) %*% r, "I")
```

```
[1] 9.689703e-14
```

It is close to zero because the residual vector is orthogonal to the column space of $\mathbf{X}$, which is a property of least squares, meaning that the residuals are not correlated with the predictors.

Question 2 (Gaussian likelihood for linear regression)

Continue with the data from Question 1. Suppose

$$\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n).$$

The log-likelihood, including the constant term, is

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta).$$

- a. For fixed σ^2 , explain why maximizing $\ell(\beta, \sigma^2)$ over β is equivalent to minimizing the residual sum of squares. What does this imply about the maximum-likelihood estimate of β ?

This implies that the maximum-likelihood estimate of β is the same as the least-squares estimate $\hat{\beta}$, since minimizing the residual sum of squares is equivalent to maximizing the log-likelihood with respect to β for a fixed σ^2 .

- b. For fixed β , differentiate the log-likelihood with respect to σ^2 and derive its maximum-likelihood estimate. Calculate this estimate at $\hat{\beta}$ and compare it with the unbiased estimate of σ^2 . Explain why the two denominators differ.

Calculate the maximum-likelihood estimate of σ^2 at $\hat{\beta}$:

```
sigma2_mle = sum(r^2) / n
sigma2_unbiased = sum(r^2) / (n - p - 1)
sigma2_mle
```

```
[1] 0.4574759
```

```
sigma2_unbiased
```

```
[1] 0.4866765
```

The two different estimates of σ^2 reflect the difference in degrees of freedom used in their calculations. The maximum-likelihood estimate uses all n observations, while the unbiased estimate accounts for the $p+1$ parameters estimated in the model.

c. Let

$$\mathbf{e}_{x_2} = (0, 0, 1, 0, 0, 0)^T.$$

Using the variance estimate from part b, plot

$$\ell(\hat{\beta} + t\mathbf{e}_{x_2}, \hat{\sigma}_{\text{MLE}}^2)$$

over a grid of t values from -1.25 to 1.25 . State where the maximum occurs and explain why this agrees with the least-squares result.

Plot:

```
e_x2 <- c(0, 0, 1, 0, 0, 0)

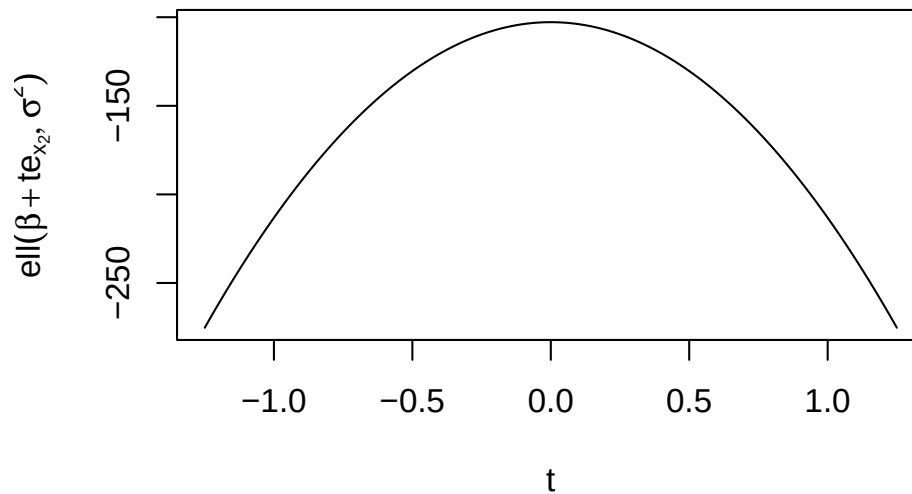
loglik <- function(beta_vec, sigma2) {
  resid <- y - X %*% beta_vec
  rss <- sum(resid^2)
  -(n/2) * log(2 * pi * sigma2) - rss / (2 * sigma2)
}

t_values <- seq(-1.25, 1.25, length.out = 500)

loglik_values <- sapply(t_values, function(t) {
  beta_t <- betahat + t * e_x2
  loglik(beta_t, sigma2_mle)
})

plot(t_values, loglik_values,
     type = "l",
     xlab = "t",
     ylab = expression(ell(beta + t*e[x[2]], sigma^2)),
     main = "Log-likelihood as a function of t")
```

Log-likelihood as a function of t



The maximum of the log-likelihood occurs at $t = 0$. This corresponds to $\hat{\beta} + 0\mathbf{e}_{x_2} = \hat{\beta}$. This agrees with the least-squares result because $\hat{\beta}$ minimizes the residual sum of squares and, for fixed σ^2 , maximizing the log-likelihood is equivalent to minimizing the residual sum of squares. Therefore, moving away from $\hat{\beta}$ in the x_2 direction decreases the log-likelihood.

Question 3 (Starting values in nonconvex optimization)

Consider the function

$$f(x) = \exp(1.5x) - 3(x + 6)^2 - 0.05x^3,$$

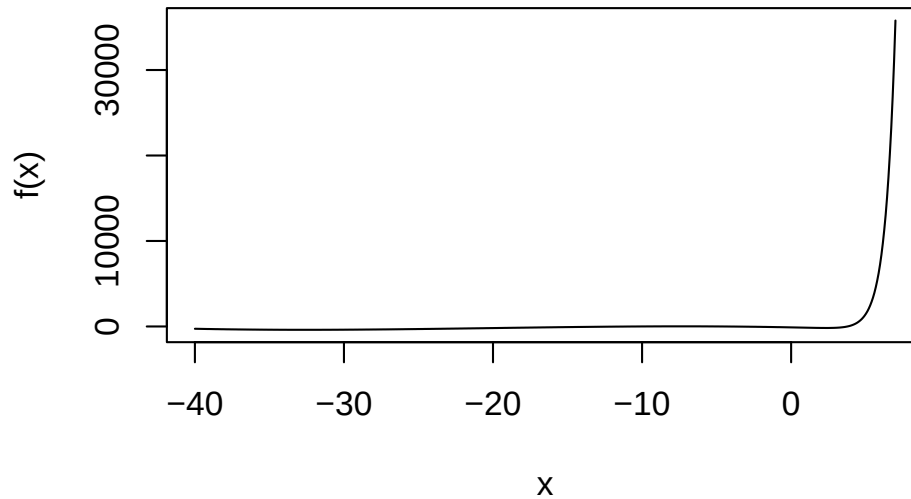
with derivative

$$f'(x) = 1.5 \exp(1.5x) - 6(x + 6) - 0.15x^2.$$

- a. Plot $f(x)$ over the interval $[-40, 7]$. Based on the plot, describe the important features of the objective function that may affect numerical optimization.

```
f <- function(x) {  
  exp(1.5 * x) - 3 * (x + 6)^2 - 0.05 * x^3  
}  
t_values <- seq(-40, 7, length.out = 500)  
  
plot(t_values, f(t_values),  
     type = "l",  
     xlab = "x",  
     ylab = "f(x)",  
     main = "Objective function")
```

Objective function



The plot shows that $f(x)$ is non-convex and has multiple local minima. There is a local minimum around $x \approx -32$ and another around $x \approx 2$. The function is relatively flat over much of the interval, but increases very rapidly as x approaches 7. These features can cause an optimizer to converge to different local minima rather than the global minimum. b. Use BFGS to minimize $f(x)$ twice, starting at $x^{(0)} = -15$ and $x^{(0)} = 0$. Supply $f'(x)$ to the optimizer. For each run, report the final value of x , the final objective value, $|f'(x)|$, and whether the optimizer reported convergence.

```
f_prime <- function(x) {
  1.5 * exp(1.5 * x) - 6 * (x + 6) - 0.15 * x^2
}

# Run 1: Starting at x^(0) = -15
result1 <- optim(par = -15, fn = f, gr = f_prime, method = "BFGS")

# Run 2: Starting at x^(0) = 0
result2 <- optim(par = 0, fn = f, gr = f_prime, method = "BFGS")
#the final value of X
result1$par
```

```
[1] -32.64911
```

```
result2$par
```

```
[1] 2.349967
```

```
#the final objective value
result1$value
```

```
[1] -390.3858
```

```
result2$value
```

```
[1] -175.8626
```

```
#the absolute value of the gradient at the final point
abs(f_prime(result1$par))
```

```
[1] 8.725087e-08
```

```
abs(f_prime(result2$par))
```

```
[1] 6.556057e-07
```

```
#convergence status
result1$convergence
```

```
[1] 0
```

```
result2$convergence
```

```
[1] 0
```

- c. Explain why the two runs can converge to different answers. Which run gives the lower objective value? What additional evidence would be needed before claiming that this point is the global minimum?

Because $f(x)$ is not necessarily convex and may have multiple local minima. BFGS is a local optimization method, so the final answer can depend on the starting value. The run starting at $x^{(0)} = -15$ gives the lower objective value, with $f(x) = -390.3858$, compared to $f(x) = -175.8626$ for the run starting at $x^{(0)} = 0$. Additional evidence, such as checking the function over a wide range of x values or using multiple starting points to determine whether there are any other points with a lower objective value.

Question 4 (Nearly collinear predictors)

Continue with $\mathbf{X}$ and $\mathbf{y}$ from Question 1. Set the random seed to 43202 and generate

$$u_i \stackrel{\text{iid}}{\sim} N(0, 1), \quad i = 1, \dots, n.$$

Define

$$\mathbf{x}_6 = \mathbf{x}_1 + 10^{-4}\mathbf{u}, \quad \mathbf{X}_+ = [\mathbf{X} \quad \mathbf{x}_6],$$

and let

$$\mathbf{y}^* = \mathbf{y} + 10^{-3}\mathbf{u}.$$

Use a numerically stable least-squares routine throughout.

- a. Fit $\mathbf{y}$ using $\mathbf{X}$ and $\mathbf{X}_+$. For each design matrix, report the condition number and training RMSE. For the augmented fit, also report the coefficients of $\mathbf{x}_1$ and $\mathbf{x}_6$.

```
set.seed(43202)
u <- rnorm(n)
x6 <- X[, 2] + 1e-4 * u
X_plus <- cbind(X, x6)
betahat_X <- qr.solve(X, y)
betahat_Xplus <- qr.solve(X_plus, y)
kappa(X)
```

```
[1] 1.950478
```

```
kappa(X_plus)
```

```
[1] 31344.8
```

```
r_X <- y - X %*% betahat_X
rmse_X <- sqrt(mean(r_X^2))

r_Xplus <- y - X_plus %*% betahat_Xplus
rmse_Xplus <- sqrt(mean(r_Xplus^2))
rmse_X
```

```
[1] 0.6763696
```

```
rmse_Xplus
```

```
[1] 0.6754668
```

```
#coefficients of x1 and x6
betahat_Xplus[2] # coefficient of x1
```

```
[1] 351.6766
```

```
betahat_Xplus[7] # coefficient of x6
```

```
[1] -350.2588
```

- b. Fit $\mathbf{y}^*$ using $\mathbf{X}_+$. Report the changes in the coefficients of $\mathbf{x}_1$ and $\mathbf{x}_6$. Compare the two fitted-value vectors using their root mean squared difference and maximum absolute difference.

```
ystar <- y + 1e-3 * u

betahat_Xplus_star <- qr.solve(X_plus, ystar)

betahat_Xplus_star[2] - betahat_Xplus[2]
```

```
[1] -10
```

```
betahat_Xplus_star[7] - betahat_Xplus[7]
```

```
[1] 10
```

```
yhat_Xplus <- X_plus %*% betahat_Xplus
yhat_Xplus_star <- X_plus %*% betahat_Xplus_star

rmsd <- sqrt(mean((yhat_Xplus - yhat_Xplus_star)^2))
maxdiff <- max(abs(yhat_Xplus - yhat_Xplus_star))
rmsd
```

```
[1] 0.001015879
```

```
maxdiff
```

```
[1] 0.002572675
```

After perturbing $\mathbf{y}$ to $\mathbf{y}^*$, the coefficient of $\mathbf{x}_1$ decreases by 10, while the coefficient of $\mathbf{x}_6$ increases by 10. Thus,

$$\Delta \hat{\beta}_1 = -10, \quad \Delta \hat{\beta}_6 = 10.$$

The root mean squared difference between the two fitted-value vectors is

```
0.001015879,
```

and the maximum absolute difference is

```
0.002572675.
```

c. Use the identity

$$\mathbf{y}^* - \mathbf{y} = 10(\mathbf{x}_6 - \mathbf{x}_1)$$

to explain why the fitted values change very little while the two coefficients change substantially. What does this example suggest about interpreting separate effects for nearly collinear predictors?

Since $\mathbf{y}^* - \mathbf{y} = 10(\mathbf{x}_6 - \mathbf{x}_1)$, the change in the response is in the same direction as the difference between the two nearly collinear predictors. Therefore, the change can be explained by increasing the coefficient of $\mathbf{x}_6$ by 10 and decreasing the coefficient of $\mathbf{x}_1$ by 10. Because $\mathbf{x}_1$ and $\mathbf{x}_6$ are very similar, these changes largely cancel each other out when calculating the fitted values, so the fitted values change very little.

This example shows that when predictors are nearly collinear, the separate coefficients can be highly unstable and difficult to interpret. ## Question 5 (Data preparation and summary statistics)

The file `data/data-manipulation.csv` is a small constructed data table containing an observation identifier, a group label, two numeric predictors, and a response. Some response values are missing. For analyses involving the response, use only observations with a recorded response. Do not replace a missing response with zero or a group mean.

- Read the data and report its dimensions, column types, number of duplicated identifiers, and number of missing values in each column. Create an analysis table containing only observations with a recorded response, and define

```
feature_sum = x1 + x2.
```

```
data <- read.csv("data/data-manipulation.csv")
```

```
dim(data)
```

```
[1] 24 5
```

```
str(data)
```

```
'data.frame': 24 obs. of 5 variables:
 $ observation_id: chr "obs-01" "obs-02" "obs-03" "obs-04" ...
 $ group         : chr "A" "A" "A" "A" ...
 $ x1            : num 1.2 2.4 0.8 3.1 2 1.7 2.8 1 3.5 2.7 ...
 $ x2            : num 2 1.1 2.8 1.5 2.2 1.8 0.9 2.5 1.4 2.3 ...
 $ response      : num 5.36 6.18 4.62 NA 5.52 5.64 7.18 4.7 7.84 7.24 ...
```



```
sum(duplicated(data$observation_id))
```

```
[1] 0
```

```
colSums(is.na(data))
```

```
observation_id      group      x1      x2      response
              0          0          0          0          3
```

```
analysis <- data[!is.na(data$response), ]
```

```
analysis$feature_sum <- analysis$x1 + analysis$x2
```

- b. For each group, report the number of observations, the mean response, and the mean of `feature_sum`. State clearly which observations these summaries describe.

```
aggregate(
  cbind(response, feature_sum) ~ group,
  data = analysis,
  FUN = mean
)
```

```
group response feature_sum
1      A 5.600000    3.600000
2      B 7.602857    4.928571
3      C 7.997143    6.585714
```

```
table(analysis$group)
```

```
A B C
7 7 7
```

These summaries are based on the analysis table, which contains only observations with a recorded response. The number of observations, mean response, and mean of `feature_sum` are calculated for each group using only the non-missing response values.

- c. Sort the analysis table by response from largest to smallest and report the first three rows, including `observation_id`, `group`, `response`, and `feature_sum`. Add code checks verifying that the number of retained rows equals the number of nonmissing responses in the original data, that the analysis table has no missing response values, and that its identifiers are unique.

```
analysis_sorted <- analysis[order(-analysis$response), ]
analysis_sorted[1:3,
  c("observation_id", "group", "response", "feature_sum")]
```

```
observation_id group response feature_sum
22      obs-22    C      9.24          6.6
16      obs-16    B      9.00          5.2
14      obs-14    B      8.66          5.0
```

```
nrow(analysis) == sum(!is.na(data$response))
```

```
[1] TRUE
```

```
sum(is.na(analysis$response)) == 0
```

```
[1] TRUE
```

```
length(unique(analysis$observation_id)) == nrow(analysis)
```

```
[1] TRUE
```

Question 6 (Can 39 million Fitbit records represent US adults?)

Patten et al. (2026) describe Fitbit data from 59,018 participants in the All of Us Research Program. The dataset spans 14 years and contains more than 39 million daily step records. Participants contributed data through one of two routes:

- Bring Your Own Device (BYOD): participants shared data from a Fitbit they already owned.
- Wearables Enhancing All of Us Research (WEAR): invited participants received a Fitbit at no cost.

In the general activity cohort, the BYOD and WEAR groups contained 32,035 and 22,474 participants, respectively. The BYOD value is listed first in each comparison below:

- 77.3% versus 55.1% reported being White;
- 6.1% versus 15.2% reported annual household income between \$10,000 and \$25,000; and
- median daily steps were 6,867 versus 5,797.

Suppose the target is the mean of the participant-specific average daily step counts among US adults during the study period, with each adult given equal weight. A researcher writes:

“This dataset contains more than 39 million daily Fitbit records. Therefore, the average of all recorded step counts should provide an accurate estimate of mean daily activity among US adults.”

- a. Are the 39 million daily records independent observations? What characteristics or behaviors could cause some participants to contribute more recorded days than others? Explain how averaging all recorded days would then weight participants unequally.

No they are not because the records are varying within participants. Some participants may forget to wear their Fitbit on some days, while others may remember all the days. Additionally, some people may have more active days due to work or lifestyle, while others are more sedentary. Therefore, averaging all recorded days would be unequal since participants with more recorded days have a greater influence.

- b. BYOD participants reported a median of 6,867 daily steps, compared with 5,797 among WEAR participants. Does this comparison show that already owning a Fitbit causes people to walk more? Explain your answer and give at least one plausible alternative explanation based on how participants entered the two groups.

No, the difference could be due to selection bias, because BYOD participants may be more active or health conscious than WEAR. For example people who already own a Fitbit may track their steps more as a habit while WEAR might be less.

- c. A very large dataset can still produce a biased estimate of a population quantity when the people and observations entering the dataset are selected. Using this study, explain how selection of both participants and recorded days could make the observed data differ from the US adult population. Why might the naive average of all recorded daily step counts fail to estimate the stated target? You do not need to determine the direction of the bias.

It gives the same weight to all recorded days, so participants with more recorded days have a greater influence on the average. Additionally, the participants in this study are not representative of the US adult population, as they are self-selected and may differ in terms of demographics, health behaviors, and other characteristics.